225947: altered volcaniclastic rock, Silvergrass prospect

(Pyradie Formation, Fortescue Group, Fortescue Basin, Capricorn Orogen)

Location and sampling

Mount Bruce (SF 50-11), McRae (2453)
MGA Zone 50, 587738E 7535155N

Sampled on 13 February 2017

This sample was collected from the 1230.5 – 1230.65 m depth interval of drillcore DD85SGS1, drilled in 1985 by CRA Exploration Pty Ltd at their Silvergrass prospect (Meakins, 1988). The drillhole is located on Hamersley Station, about 5.3 km north-northwest of Four Corners Bore, 4.7 km east-southeast of Kangeenareena Well, and 1.9 km east of Cattle Well.

Geological context

The unit sampled is the 2730–2716 Ma Pyradie Formation of sequence 4 of the Fortescue Group in the South Pilbara Sub-basin (Thorne and Trendall, 2001; Howard, 2021b). The formation in general is up to 1.2 km thick (365 m in drillcore DD85SGS1), and consists mainly of basalt flows and pillow lavas, tuffaceous and volcaniclastic rocks, hyaloclastite, minor komatiite and chert. In this drillcore, the Pyradie Formation is dominated by tuffaceous and volcaniclastic rocks, siliciclastic sedimentary rocks and there are only minor basalt flows and pillow lavas (Meakins, 1988). The Pyradie Formation conformably overlies the 2745–2730 Ma Boongal Formation and is conformably overlain by the 2718–2715 Ma Bunjinah Formation (Thorne and Trendall, 2001; Howard, 2021a,b). A dolerite sill intruded into Hardey Formation at Coppin Pool, about 73 km to the south-southeast, yielded an igneous crystallization age of 2716 ± 5 Ma (GSWA 228400, Wingate et al., 2020), and is correlated with the Pyradie Formation based on distinctive geochemistry (Howard, in prep.).

Petrographic description

The sample is interpreted as a very fine-grained volcaniclastic rock based on visual examination of drillcore (Fig. 1). Microscopic examination indicates the sample is very highly altered to clinozoisite and quartz, and does not exhibit primary mineralogy or diagnostic relict textures. A subtle planar lamination is defined by a series of parallel, discontinuous layers up to 1 mm wide of secondary quartz and by higher concentrations of fine-grained titanite, probably after original very fine-grained oxide minerals. The laminations are crosscut by a network of quartz veins up to 1 mm wide, some containing minor carbonate minerals. Large areas of the thin section consist almost entirely of massive, fine-grained colourless clinozoisite, which forms a mosaic of anhedral, mainly equant grains up to 0.1 mm in size. Secondary quartz forms isolated small grains within the clinozoisite mosaic but mainly forms aggregates of <0.1 mm anhedral grains in the quartz-replaced narrow laminae. Dispersed titanite grains, probably after original magnetite, make up <1% of the sample. Within the fine-grained clinozoisite is a 10 mm-wide band that contains dispersed, equant to elongate grain shapes of unknown origin composed of clinozoisite with minor carbonate and quartz. The sample contains rare disseminated chalcopyrite grains up to 0.05 mm in size. The rock is not obviously deformed or metamorphosed, although its origin is masked by the intense alteration.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and anhedral to euhedral. The crystals are up to 150 µm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A few strongly rounded zircons exhibit concentric zoning truncated at grain margins, consistent with abrasion during sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 23–24 October 2020, using SHRIMP-B. Eleven analyses of the M257 standard were obtained during the session, of which 10 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.21% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Seventeen analyses were obtained from 17 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

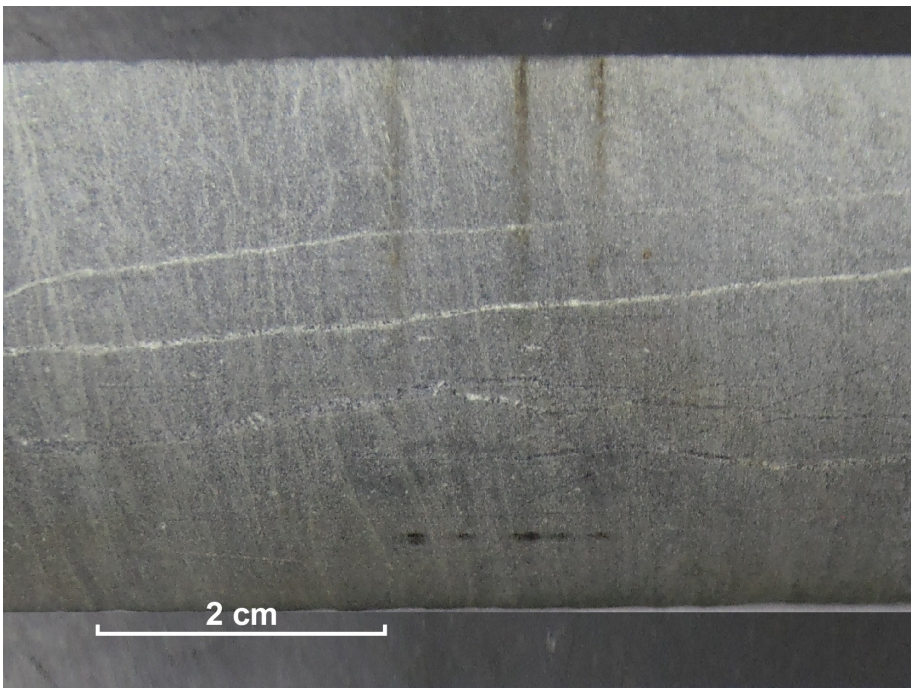

Figure 1. **Drillcore image for sample 225947: altered volcaniclastic rock, Silvergrass prospect**

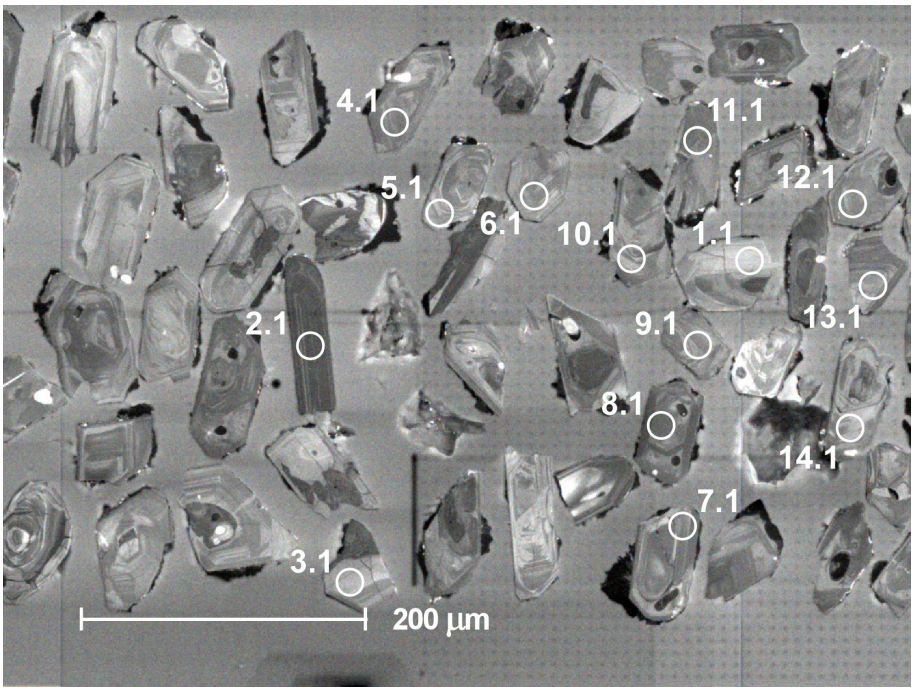

Figure 2. **Cathodoluminescence image of representative zircons from sample 225947: altered volcaniclastic rock, Silvergrass prospect. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 225947: altered volcaniclastic rock, Silvergrass prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	1	1.1	69	33	0.48	0.410	1.930	0.034	0.18858	0.00198	1.938	0.034	0.18492	0.00237	2682	39	2698	21	0.6
I	4	4.1	131	96	0.73	-0.052	1.969	0.026	0.18539	0.00230	1.968	0.026	0.18585	0.00233	2648	29	2706	21	2.1
I	3	3.1	68	33	0.49	0.053	1.938	0.035	0.18698	0.00201	1.939	0.035	0.18651	0.00207	2681	40	2712	18	1.1
I	5	5.1	101	60	0.60	-0.033	1.949	0.029	0.18649	0.00159	1.948	0.029	0.18678	0.00162	2671	33	2714	14	1.6
I	15	15.1	147	87	0.59	-0.055	1.948	0.026	0.18662	0.00146	1.947	0.026	0.18710	0.00150	2673	30	2717	13	1.6
I	9	9.1	108	58	0.53	0.000	1.882	0.028	0.18792	0.00161	1.882	0.028	0.18792	0.00161	2748	34	2724	14	-0.9
I	8	8.1	167	141	0.84	0.020	1.907	0.023	0.18813	0.00126	1.908	0.023	0.18795	0.00127	2717	27	2724	11	0.3
I	11	11.1	120	72	0.60	-0.025	1.960	0.026	0.18784	0.00139	1.959	0.026	0.18806	0.00141	2658	29	2725	12	2.5
I	10	10.1	128	68	0.54	0.081	1.989	0.027	0.18890	0.00147	1.991	0.027	0.18817	0.00153	2624	30	2726	13	3.8
I	13	13.1	104	63	0.60	0.142	1.935	0.029	0.18980	0.00167	1.938	0.029	0.18854	0.00179	2683	34	2729	16	1.7
I	7	7.1	166	121	0.73	0.042	1.949	0.024	0.18908	0.00128	1.950	0.024	0.18871	0.00131	2669	27	2731	11	2.3
I	14	14.1	80	41	0.51	0.093	1.888	0.033	0.19044	0.00192	1.890	0.033	0.18962	0.00201	2738	39	2739	17	0.0
I	6	6.1	107	68	0.63	-0.136	1.890	0.028	0.18858	0.00163	1.887	0.028	0.18979	0.00174	2741	33	2740	15	0.0
I	16	16.1	86	41	0.48	-0.044	1.864	0.031	0.18994	0.00186	1.864	0.031	0.19034	0.00190	2769	38	2745	16	-0.9
I	12	12.1	112	64	0.57	-0.314	1.909	0.049	0.18797	0.00157	1.903	0.049	0.19077	0.00179	2723	57	2749	15	0.9
D	2	2.1	1367	543	0.40	0.004	3.299	0.025	0.11311	0.00041	3.299	0.025	0.11308	0.00041	1707	12	1849	7	7.7
D	17	17.1	168	113	0.67	0.053	2.133	0.028	0.18208	0.00139	2.134	0.028	0.18160	0.00144	2478	27	2668	13	7.1

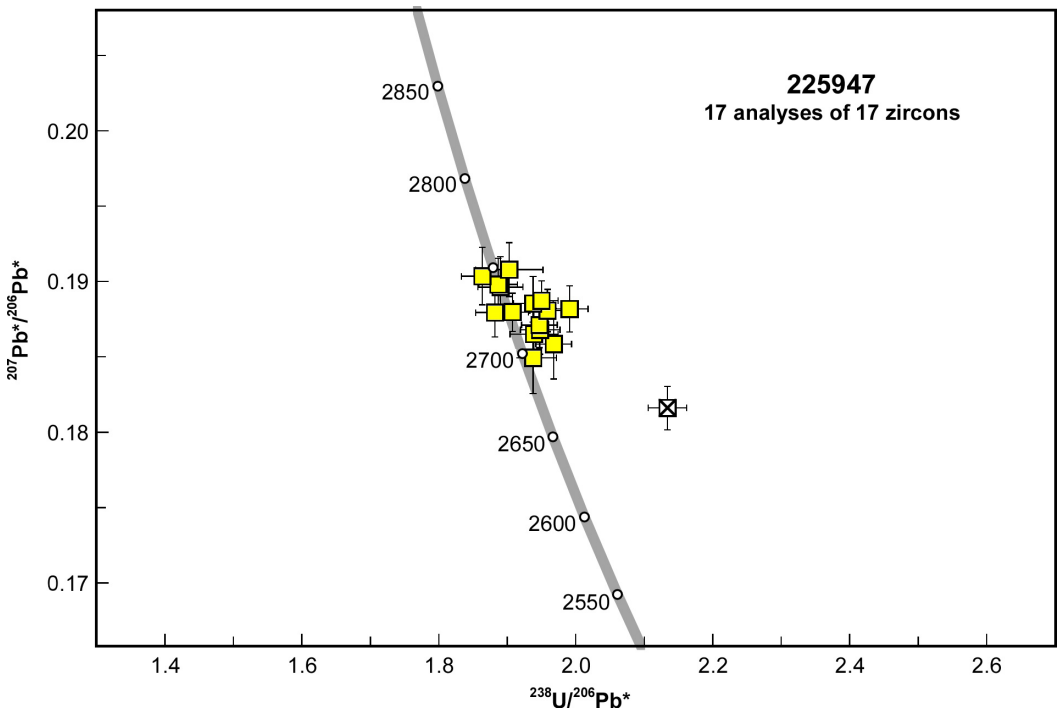

Figure 3. U–Pb analytical data for zircons from sample 225947: altered volcaniclastic rock, Silvergrass prospect. Yellow squares indicate Group I (magmatic zircons); crossed square indicates Group D (discordance >5%). One analysis in Group D is not shown

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 15 analyses form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 15 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2727 ± 7 Ma (MSWD = 0.70).

The date of 2727 ± 7 Ma for the 15 analyses in Group I is interpreted as the igneous crystallization age of the volcanic rock.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Howard, HM 2021, 225947: altered volcaniclastic rock, Silvergrass prospect; Geochronology Record 1816: Geological Survey of Western Australia, 4p.

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Data released: 12 November 2021